

TECHNICAL REPORT



NHS Appointment Utilisation & Missed Appointments

Analysis of NHS appointment data to identify capacity utilisation patterns, missed appointment drivers, and data quality constraints across 106 locations and 7 regions.

Andrew Willacy | August 2025

TOOLS & TECH STACK



Excel

Data preparation & EDA



Python

Analysis



matplotlib

Matplotlib / Seaborn

Charts & dashboards

KEY BUSINESS QUESTIONS

Is capacity adequate?

Staff and network vs demand

How are resources used?

Actual appointment utilisation

What drives missed appts?

DNA rates, patterns, and costs

NHS APPOINTMENT UTILISATION & MISSED APPOINTMENTS ANALYSIS

Technical Report — Andrew Willacy — August 2025

Executive Summary

This report presents the findings of an analysis of NHS appointment data, examining capacity utilisation, missed appointment patterns, and resource deployment across primary care in England. Using Python for analysis and visualisation, the work addressed three core questions: whether adequate capacity has been in place, how resources have actually been utilised, and what is driving missed appointments.

The central finding is not a simple answer to these questions, but a structural constraint on answering them at all: data quality is the primary barrier to understanding NHS performance. Unmapped and unknown values consistently dominate key metrics, the daily capacity benchmark is fundamentally misconfigured, and the Twitter dataset is insufficiently structured for meaningful sentiment analysis in its current form.

Finding	Summary
Scale of dataset	Over 1 million appointment records across 106 locations and 7 regions
Service composition	General practice accounts for 91.5% of all appointments
Missed appointments	4–5% DNA rate across the period; a further 4–5% unmapped
Capacity benchmark	Fixed 1.2M daily capacity figure is applied uniformly including weekends — fundamentally skewing utilisation analysis
Data quality	Unknown/data quality category is frequently the largest count in key metrics — the primary barrier to reliable insight
Twitter dataset	Insufficient structure for meaningful analysis in current form

Business Problem

The NHS faces increasing demand, constrained resources, and a persistent missed appointment problem. With an ageing population placing growing strain on primary care, understanding whether existing capacity is adequate — and why appointments are missed — is operationally and financially critical.

Each missed appointment carries an estimated cost of £30, and represents a lost care opportunity for another patient. The analysis was commissioned to address two core questions:

- Has there been adequate staff and network capacity?
- What is the actual utilisation of resources?

A secondary objective was to assess whether Twitter data could contribute meaningfully to understanding public sentiment around NHS services, and whether this could inform patient engagement strategies.

Data & Analytical Approach

Datasets

Four datasets were used for the analysis:

Dataset	Key Fields	Purpose
actual_duration.csv	Appointment durations, location, date, count	Duration and volume analysis
appointments_regional.csv	Mode, date, location, lead time, count	Regional utilisation and booking patterns
national_categories.xlsx	Treatment category, date, location, count	Service type and urgency trends
twitter.csv	Tweets referencing NHS topics	Public sentiment (limited applicability)

The Excel file (national_categories.xlsx) was converted to CSV for consistency before import. All datasets were checked for missing values and duplicates. Duplicates were found only in the appointments_regional dataset and removed. Cleaned datasets were saved with a _clean suffix.

Analytical Approach

Python was the primary analytical tool, using Pandas for data manipulation and Matplotlib and Seaborn for visualisation. Analysis was structured around the two core business questions, with an additional review of the Twitter dataset for potential sentiment application.

Key analytical decisions:

- The 1.2 million daily capacity benchmark was used as provided, but identified as a material limitation: it is applied uniformly across all days including weekends, where appointment volumes are significantly lower. This is flagged throughout where it affects interpretation.
- Monday appointment counts were noted as potentially inflated due to bank holiday displacement — a further distortion in day-of-week analysis.
- The Twitter dataset was reviewed for trending hashtags and keyword patterns only; full sentiment analysis was not feasible given the dataset's current structure.

Key Findings

1. Appointment Volumes and Composition

The dataset spans over one million appointment records across 106 locations and seven NHS regions. General practice dominates the service landscape, accounting for 91.5% of all appointments — meaning patterns identified in the data are overwhelmingly reflective of GP activity.

Metric	Finding
Total locations	106 across 7 regions
Dominant service type	General practice — 91.5% of all appointments
Peak month (volume)	November 2021 — 30.4 million appointments
Lowest month (volume)	August 2021 — 23.9 million appointments
Most common mode	Face-to-face; telephone became more prevalent during Covid
Most common duration	1–5 minutes most frequent by count; 6–10 minutes typical

Seasonal patterns are visible: appointment volumes dip in winter months (December–February), likely reflecting a combination of holiday periods and weather impacts. This is a material consideration for capacity planning, as staffing decisions should reflect these fluctuations rather than a fixed daily benchmark.

A significant data quality issue affects duration analysis: the 'Unknown / data quality' category accounts for the largest count in duration metrics — larger than any specific duration band. This limits the reliability of duration-based conclusions and is the most significant single data gap in the dataset.

2. Capacity Utilisation

The NHS uses a benchmark of 1.2 million appointments per day to assess capacity. Applied to the dataset, this produces apparent utilisation figures — but the benchmark is fundamentally misconfigured for meaningful analysis.

- The 1.2 million figure is applied uniformly to every day, including weekends and bank holidays, when appointment volumes are a fraction of weekday levels. This produces artificially low utilisation rates on weekdays and creates a misleading picture of overall capacity deployment.
- Monday counts are further distorted by bank holiday displacement: appointments that would have been scheduled on a bank holiday appear to shift to Monday, inflating that day's figures.
- The appropriate comparison is not appointments against a fixed daily ceiling, but appointments against available scheduled capacity by day type — a distinction the current data structure does not support without additional metadata.

The broader picture suggests increasing demand pressure, particularly in general practice. A steady upward trend in total appointments is visible over the 2020–2022 period, interrupted by Covid lockdowns before recovering. Evidence of infrastructure strain appears during peak periods — winter months and flu seasons — where demand concentrates while capacity remains fixed.

3. Missed Appointments

Missed appointments (DNA — Did Not Attend) represent a persistent and quantifiable cost to the NHS, both financially and in terms of care capacity lost to other patients.

Metric	Finding
DNA rate	4–5% consistently across the period
Unmapped appointments	A further 4–5% with no status classification
Appointment type pattern	Missed appointments more frequent in routine than urgent appointments
Estimated cost per missed appt	£30 per appointment

The 4–5% DNA rate is consistent across the period. The additional 4–5% unmapped category is a compounding concern: without knowing the outcome of these appointments, neither utilisation nor DNA analysis can be considered complete. The gap between attended, DNA, and unclassified appointments represents a fundamental data collection problem rather than an analytical one.

The pattern of routine appointments generating more missed slots than urgent ones is operationally logical — patients may be more likely to deprioritise non-urgent appointments — but also points to a specific intervention opportunity: routine appointment reminders, with friction-free rescheduling options, are likely to have a measurable impact.

4. Twitter / Social Media Data

The Twitter dataset was reviewed for hashtag trends and keyword patterns as a potential input into public sentiment analysis. The most frequent hashtags included #healthcare, #health, and #medicine.

However, the dataset in its current form has limited analytical value. It lacks the structure required for reliable sentiment classification: volume is insufficient for regional segmentation, metadata is sparse, and the hashtag-level analysis available offers directional signals only. Any conclusions drawn from the Twitter data in its current state would carry low confidence.

This is not a reason to abandon social media as an input; it is a reason to invest in better data collection if social listening is a genuine strategic priority.

Recommendations

1. Fix the Capacity Benchmark

- Replace the uniform 1.2 million daily capacity figure with a demand-weighted model that reflects actual scheduled availability by day type (weekday, Saturday, Sunday, bank holiday).
- Model future demand using historical trend data, with explicit recognition that the 2020–2022 period is atypical due to Covid disruption and should not be treated as a stable baseline.
- Align capacity planning to seasonal demand patterns — prioritising face-to-face and urgent care resourcing during winter peaks, as a retailer would increase staffing during high-demand trading periods.

2. Resolve the Data Quality Problem

- The 'Unknown / data quality' category must be addressed at source. Until it is, appointment duration, utilisation, and status analysis will remain unreliable. This is the single highest-priority action from this analysis.
- Standardise data collection processes across all 106 locations to reduce unmapped and unclassified entries. Accurate data is a prerequisite for accurate insight; analytical sophistication cannot compensate for structural data gaps.
- Clarify and document what the 1.2 million daily capacity benchmark represents, how it is derived, and under what conditions it should be updated.

3. Reduce Missed Appointments

- Implement SMS or email reminders for routine appointments — the segment with the highest DNA rate. Acknowledge that not all patients are digitally engaged and ensure non-digital alternatives remain available.
- Reduce friction in rescheduling: patient portals and apps that allow appointment management without requiring a phone call to reception are likely to reduce avoidable DNAs.
- Investigate regional and demographic patterns in missed appointments once the unmapped category is resolved — targeted awareness campaigns can then be grounded in evidence.

4. Invest in Social Media Data Infrastructure

- If social listening is a strategic objective, commission a structured data collection approach rather than relying on ad hoc Twitter datasets. Volume, geographic tagging, and consistent metadata are prerequisites for reliable sentiment analysis.
- Develop a regular dashboard to monitor sentiment trends across major regions, drawing on multiple platforms (not Twitter alone).

- Consider using social media proactively as a communication channel — particularly for appointment reminders, campaign awareness, and public health messaging during crisis periods.

Limitations

Limitation	Impact	Mitigation Applied
Fixed daily capacity benchmark (1.2M)	Distorts utilisation rates — particularly on weekends and bank holidays	Flagged throughout; conclusions on utilisation treated as directional only
Unknown / data quality category	Largest count in key metrics — limits reliability of duration and status analysis	Identified as the primary data constraint; findings scoped accordingly
Unmapped appointment status (4–5%)	DNA and attended rates cannot be fully reconciled	Reported separately; not included in DNA calculations
Atypical time period (2020–2022)	Covid disruption means trends are unlikely to represent normal operating conditions	Explicitly noted; findings treated as period-specific, not generalisable
Monday displacement effect	Bank holiday rescheduling inflates Monday appointment counts	Flagged in day-of-week analysis
Twitter dataset quality	Insufficient structure and volume for reliable sentiment analysis	Reviewed for directional signals only; conclusions treated as indicative
No demographic data	Cannot identify which patient groups drive missed appointments	Highlighted as a key gap; demographic data collection recommended

Further Analysis

The following extensions would materially improve the analytical value of this work:

- Demand-weighted capacity modelling — replace the fixed 1.2M benchmark with a model that reflects actual scheduled capacity by day type, location, and season. This is the most impactful near-term analytical extension.
- Demographic integration — incorporating patient demographic and socioeconomic data alongside appointment records would enable identification of which groups are most likely to miss appointments and allow targeted intervention design.
- Longitudinal trend analysis — extending the dataset beyond 2022 would allow separation of Covid-era disruption from underlying structural trends, producing a more reliable baseline for future capacity planning.

- Multi-channel social media analysis — structured data collection across Twitter, Facebook, and other platforms with consistent metadata would enable reliable sentiment analysis, regional segmentation, and real-time public communication monitoring.
- Predictive DNA modelling — with demographic data added, a classification model could be built to identify appointments at elevated risk of being missed, enabling proactive reminder targeting and resource reallocation.
- Transport and contextual factor analysis — incorporating transport accessibility, weather, and employment patterns as external variables could improve understanding of the structural drivers of missed appointments beyond patient behaviour alone.

Repository Structure

Project files are available at: <https://github.com/AndrewWillacy/NHS-Appointment-Utilisation-and-Missed-Appointments-Analysis>

File / Folder	Contents
actual_duration_clean.csv	Cleaned appointment duration dataset
appointments_regional_clean.csv	Cleaned regional appointments dataset (duplicates removed)
national_categories_clean.csv	Cleaned national categories dataset (converted from Excel)
NHS_Analysis_Notebook.ipynb	Python analysis notebook with all visualisations and commentary
Technical_Report.pdf	This document
metadata.txt	Field descriptions for all dataset columns
